

GEBZE TECHNICAL UNIVERSITY PHYSICS DEPARTMENT

PHYS 424 : CLASSICAL MECHANICS II

2025 Spring Semester, Homework II (Monday, May 12, 2025)

INSTRUCTIONS :

1. You cannot ask someone to show you how to completely solve the problem, nor can you accept such extensive help.
 2. Under no circumstances should you copy or even look at someone's solutions.
 3. Write down your solutions clearly and readable. Submit them on the due date during the lecture.
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1) A comet of mass 1.2×10^{10} kg moves in an elliptical orbit around the Sun. Its distance from the Sun ranges between 0.5 AU and 50 AU. Note that the astronomical unit (AU) is the average distance from Sun to Earth = 149.6 million km.

- a) What is the eccentricity of its orbit ?
- b) What is its period ?
- c) At aphelion what is the potential energy of the comet-Sun system ?

2) Show that in an elliptical orbit, square of the period of an orbit, T , is proportional to the cube of the semimajor-axis length, a .

3) The Moon moves in a nearly circular orbit of radius 384,000 km and period 27.32 days. Callisto, the fourth moon of Jupiter, moves in a nearly circular orbit of radius 1,883,000 km and period 16.69 days. Estimate the mass of Jupiter as a multiple of the mass of the Earth.

4) Answer the following questions.

- a) Extremely high tides occur when the Moon is full and when the Moon is new. Why ?
- b) The tides generated by the Sun are only about half as large as the tides generated by the Moon. Explain should this make sense.

5) Calculate the amount of work required to move an Earth satellite (mass m) from a circular orbit of radius $2R_E$ to one of radius $3R_E$.

6) If a particle is projected vertically upward to a height h above a point on Earth's surface at a northern latitude λ , show that it strikes the ground at a point $\frac{4}{3}\omega \cos \lambda \sqrt{\frac{8h^3}{g}}$ to the west. (Neglect air resistance, and consider only small vertical heights.)